

Flow A - AWS Identity Triage with TheHive Alert Creation

Real-time triage, enrichment, scoring, analyst notification, and case-ready alert creation for AWS identity findings

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Flow A AWS Identity Triage TheHive Alert

Document Type

Workflow project document | Flow A

Prototype Focus

AWS IAM automation | n8n | TheHive 5 | Slack

Project Character

Portfolio prototype with implementation, testing, and validation evidence

Design Goal

Showcase an end-to-end SOC automation workflow with clear business and analyst value

Why this document matters

- This write-up is designed to work both as a project portfolio document and as an implementation-facing prototype note.
- It explains the problem, the workflow design, the operational value to a SOC, and the validation evidence that proves the workflow worked in the lab.
- The screenshots included throughout the document are the actual workflow, node, Slack, DataTable, AWS, and TheHive validation artifacts collected during the build process.

Key outcomes

- Normalizes inbound AWS identity findings delivered through SNS and n8n webhook ingestion.
- Enriches access-key findings with IAM last-used context and CloudTrail activity when available.
- Calculates a practical triage score, pushes the result to Slack and DataTable, and creates a TheHive 5 alert for analyst tracking.

1. Project Summary

Flow A is the entry-point detection and triage workflow of the prototype. It takes AWS identity findings - especially GuardDuty or imported identity-related findings - and transforms them into analyst-ready security context instead of forwarding a raw event blob.

The workflow was built to solve a very common SOC pain point: a security finding often arrives faster than an analyst can manually validate the key details, enrich the event with identity context, and decide whether it deserves further investigation. Flow A reduces that early-response friction by packaging the finding into a structured, human-readable result that can be used immediately in Slack, DataTable, and TheHive.

2. Problem Statement

AWS identity findings can be noisy, incomplete, or missing analyst-friendly context at the point of detection. A raw finding does not automatically tell an analyst whether the affected access key is valid, when it was last used, which service was involved, whether CloudTrail shows nearby events, or how urgently the case should be handled. Without automation, analysts spend valuable time manually pivoting through IAM, CloudTrail, spreadsheets, chat updates, and case tools before they can even begin response.

Why the problem matters in a SOC

- It increases early-stage alert handling time because analysts must enrich the event before making a decision.
- It contributes to alert fatigue because every finding is not equally important, but raw events often look equally urgent.
- It creates inconsistency because different analysts enrich and summarize findings in different ways.
- It delays case creation because investigation platforms like TheHive work best when the incoming alert is already structured.

3. Objective

- Receive and validate AWS SNS-driven security notifications in n8n.
- Normalize identity-related events into a consistent internal structure.
- Enrich supported access-key findings with IAM and CloudTrail context.
- Calculate a practical triage score that is easy to interpret and explain.
- Notify analysts through Slack, record the finding in DataTable, and create a matching TheHive alert.

4. Solution Overview

The workflow starts with webhook-driven SNS intake, validates the trusted topic, and handles both subscription confirmation and notification delivery. After normalization, the workflow checks whether the event is identity-related and whether an access key is available and properly formatted for enrichment.

If the key can be enriched, the workflow queries IAM for last-used details and CloudTrail for nearby management events. If enrichment cannot be completed, the workflow still preserves the original finding and marks the enrichment state clearly so the analyst understands what was and was not available.

The result is then scored, formatted for Slack, upserted to DataTable for state tracking, and pushed into TheHive 5 as an external alert. This makes Flow A both a triage workflow and the opening record-creation step for later response and case management.

5. Tools and Technologies Used

Tool / Technology	Why it was used in this workflow
AWS SNS / GuardDuty / Security Hub imported findings	Provide the upstream cloud security detection signal that starts the workflow.
n8n	Acts as the orchestration layer for validation, branching, enrichment, formatting, and downstream integration.
AWS IAM API	Used to retrieve access-key last-used context so the finding contains real identity activity information.
AWS CloudTrail LookupEvents	Used to pull adjacent management-event context for investigation when trail coverage exists.
Slack	Delivers an analyst-readable triage message with the final score and enrichment result.
n8n DataTable	Stores a persistent workflow record so the finding can be updated by later flows.
TheHive 5	Receives the structured triage result as an alert so the investigation can continue inside case-management workflows.

6. How the System Works

Stage	What happens
Webhook / SNS intake	Receives the event, confirms subscriptions when required, and acknowledges notifications.
Normalization	Converts the event into a consistent identity-finding object used by the rest of the workflow.
Eligibility checks	Confirms the event is identity-related and determines whether access-key enrichment is supported.
IAM / CloudTrail enrichment	Retrieves last-used details and nearby activity when the key and permissions allow it.
Risk scoring	Applies practical scoring logic so critical and enriched findings are easier to prioritize.
Output stage	Posts the result to Slack, upserts the tracking record, and creates a TheHive alert.

Plain-English view

- In plain English, Flow A takes a cloud identity alert and turns it into something an analyst can actually work with.
- Instead of reading raw AWS JSON and manually checking IAM and CloudTrail, the analyst receives a readable summary, a risk score, and a linked TheHive alert.

7. Workflow and Architecture Evidence

The first screenshots below show the final Flow A workflow as it exists in n8n. The full view provides the end-to-end orchestration, while the zoomed views make it easier to explain the left-side intake logic and the right-side enrichment-plus-output logic.



Figure A1. Full Flow A n8n workflow.

The workflow was intentionally split into zoomed sections to keep the design readable in documentation. This is useful because Flow A contains both event-handling logic and downstream integration logic.

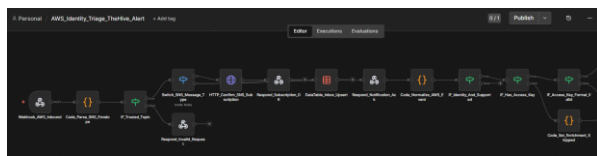


Figure A2. Flow A intake and normalization segment.

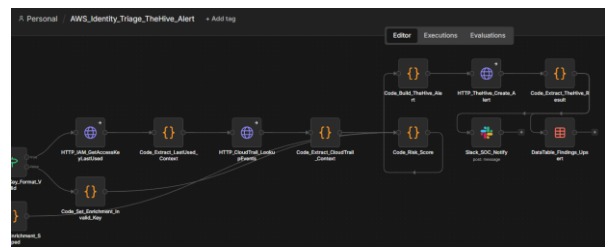


Figure A3. Flow A enrichment, scoring, Slack, Data Table, and TheHive segment.

The logic layer is important because this workflow is not only a transport pipeline; it performs validation, enrichment extraction, and scoring. The screenshots below highlight examples of the JavaScript logic and node configuration used in the working build.

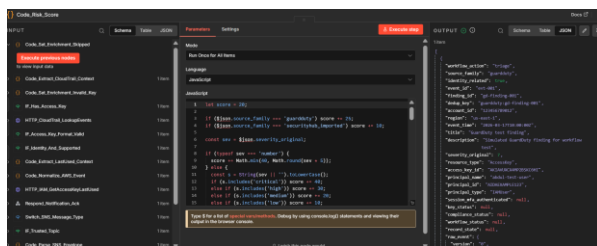


Figure A4. Scoring logic / code implementation evidence.

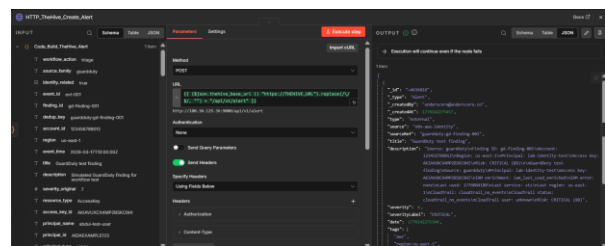


Figure A5. TheHive alert creation node configuration in the working flow.

8. Key Features and USP

- Trusted SNS topic validation before notification processing.
- Support for subscription confirmation and live notification handling in a single workflow.
- Graceful handling of unsupported or non-enrichable findings without dropping the record.
- IAM last-used enrichment and optional CloudTrail event context.
- Readable Slack message with score, identity, key, enrichment status, and TheHive linkage.
- TheHive 5 native alert creation for downstream investigation and promotion in later flows.

9. SOC Value and Expected Impact

From a SOC perspective, Flow A improves the quality of the first analyst touch-point. It does not claim to eliminate investigation, but it reduces the time needed to understand a finding, improves consistency in how findings are summarized, and gives the analyst a better-quality object to work from in TheHive. The expected operational value is lower triage delay, more consistent prioritization, and smoother escalation into containment or case work.

Operational area	Expected effect
MTTD / initial understanding	Faster interpretation because core identity context is already summarized.
MTTR	Potential reduction because later flows start from a better structured record.
Alert fatigue	Reduced by scoring and summarizing findings instead of forwarding raw telemetry.
Analyst consistency	Improved because enrichment and message formatting are standardized.
Case readiness	Improved because TheHive receives a structured, external alert immediately.

10. Validation and Testing Evidence

The validation process confirmed that Flow A successfully created a readable analyst notification in Slack. This proves the workflow did not only run in n8n; it produced a usable analyst-facing result.

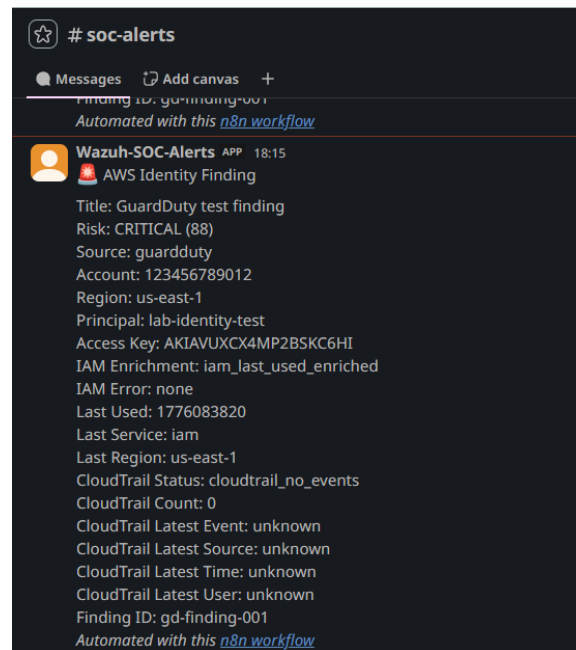


Figure A6. Slack notification produced by Flow A after triage and enrichment.

The workflow also maintained a state record in the n8n DataTable, which is important because later flows depend on a persistent finding record instead of only an ephemeral Slack post.

T_descpt_key	T_finding_id	T_source_family	T_account_id	T_region	T_event_time	T_title	T_account_key_id	T_principal_name	T_risk_bucket	T_risk_score
Null	Null	Null	Null	Null	Null	Null	Null	Null	Null	Null
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T16:30:00.000+05:30	GuardDuty test finding	AKIA_TEST_123	abul-test-user	critical	80
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T16:30:00.000+05:30	GuardDuty test finding	AKIA1234567890ABCDEF	abul-test-user	critical	80
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T16:30:00.000+05:30	GuardDuty test finding	AKIAVUXCX4MP2BSKC6HI	abul-test-user	critical	80
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T16:30:00.000+05:30	GuardDuty test finding	AKIAVUXCX4MP2BSKC6HI	abul-test-user	critical	80
guardduty-gd-finding-001	gd-finding-001	guardduty	123456789012	us-east-1	2026-03-17T16:30:00.000+05:30	GuardDuty test finding	AKIAVUXCX4MP2BSKC6HI	abul-test-user	critical	80
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Figure A7. DataTable finding record created by Flow A.

The final integration test showed that TheHive alert creation worked as expected. The screenshots below show the resulting alert list entry and the alert detail page inside TheHive 5.

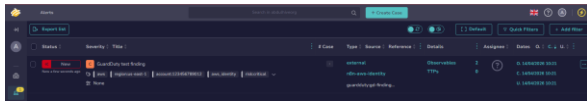


Figure A8. TheHive alert list showing the Flow A alert.

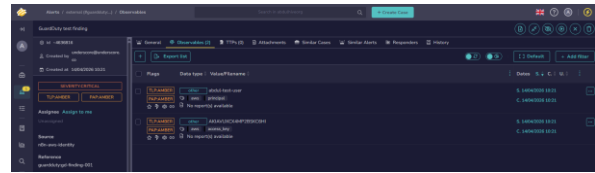


Figure A9. TheHive alert detail with observables and tags.

Once TheHive integration was added successfully, the Slack message also included the resulting TheHive identifiers and status fields. This gives the analyst an immediate cross-platform hand-off reference.

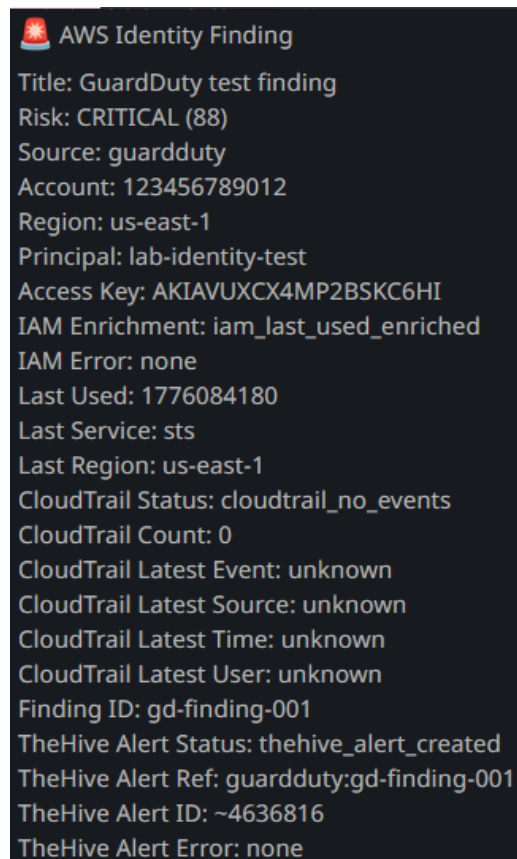


Figure A10. Final Slack validation showing TheHive alert identifiers attached to the triage result.

11. Real-World Use Cases

- GuardDuty or custom identity findings that reference an IAM access key.
- SOC teams that want a structured alert object before case escalation.
- Cloud security labs or prototypes demonstrating early-stage SOAR orchestration.

12. Skills Demonstrated

- Workflow orchestration in n8n.
- AWS event normalization and cloud security enrichment logic.

- Practical API integration across IAM, CloudTrail, Slack, TheHive, and DataTable.
- Security-focused message design for analyst consumption.
- Prototype validation and troubleshooting across multiple platforms.

13. Prototype Limitations

- The workflow depends on the upstream finding quality and may not enrich every event type equally.
- CloudTrail context depends on trail coverage and IAM permissions, so some findings may legitimately show no event evidence.
- Scoring is intentionally practical and explainable, not a statistically trained risk model.

14. Future Enhancements

- Add richer entity mapping for assumed roles, federated identities, and multi-account environments.
- Add optional analyst approval gating for certain finding classes before escalation.
- Enrich with additional context sources such as Access Analyzer, session context, or IAM policy metadata.

15. Conclusion

Flow A demonstrates the front-end intelligence layer of the prototype. It proves that an AWS identity finding can be ingested, normalized, enriched, scored, tracked, and turned into a case-ready TheHive alert with very little manual effort. That makes it a strong portfolio project on its own and an essential foundation for the rest of the prototype.

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